2nd Order DEs: Hooke's Law

Questions 1–26

Question 1. A light elastic string has natural length 1.2 m. In each case, find the extension of the string when the string

a) is stretched to a length of 1.35 m
b) is stretched to a length of 1.5 m
c) has length 1.1 m.

Question 2. A light elastic string has natural length 0.8 m and modulus of elasticity 24 N. Find the tension when the string has length

a) 1.0 m
b) 1.3 m
c) 0.75 m.

Question 3. A light elastic string has natural length 2 m. When the string is stretched to a length of 2.5 m, the tension is 8 N. Find the modulus of elasticity λ.

Question 4. A light elastic string has natural length 1.5 m and modulus of elasticity 18 N. Find the length of the string when the tension is 4.8 N. **Q5–Q6 use** $g = 9.8$ **m s**$^{-2}$**.**

Question 5. A particle of mass 0.5 kg hangs in equilibrium from a light elastic string of natural length 1.2 m and modulus of elasticity 30 N. Find the extension of the string in the equilibrium position.

Question 6. A particle of mass 0.8 kg hangs in equilibrium from a light elastic string of natural length 0.6 m. The modulus of elasticity is 19.6 N.

a) Find the extension in equilibrium.

b) Find the length of the string in equilibrium.

Question 7. A light elastic string has natural length 1 m and modulus of elasticity 40 N. One end is fixed at point A, and a particle P hangs vertically below A. Let y be the distance of P below A.

a) Show that the tension is $40(y-1)$ N when the string is taut.

b) State the value of the tension when $y < 1$.

Question 8. A particle of mass m kg hangs from a light elastic spring of natural length l m and modulus of elasticity λ. The particle is at a distance of y m vertically below the fixed point. Take downwards as positive.

a) Write down the extension of the spring.

b) Write down the tension in terms of y.

c) Assuming the spring remains taut, show that the equation of motion is

$$m\frac{d^2y}{dt^2} = mg - \lambda\frac{y-l}{l}.$$

Question 9. A particle P of mass m kg hangs from a light elastic spring of natural length l m and modulus of elasticity λ. The spring is fixed at a point A above P. The particle hangs at rest in equilibrium. Let the equilibrium extension of the spring be e m. The particle is then displaced vertically and allowed to move.



Let x be the downward displacement of the particle from its equilibrium position. Take downwards as positive.

a) Write down the length of the spring when the particle is at equilibrium.
b) When the particle is displaced a further distance x m downwards from equilibrium, write down the new length of the spring.
c) Hence show that the extension of the spring at this position is $e + x$.
d) Using Hooke's Law, write down an expression for the tension in the spring in terms of e and x.
e) When the particle is at equilibrium, explain why $mg = \lambda\frac{e}{l}$.

Question 10. A particle of mass m kg hangs from a light elastic spring of natural length l m and modulus of elasticity λ. The particle moves vertically, and the spring is always taut. The particle is initially at its equilibrium position. Let the equilibrium extension of the spring be e m. Later the particle is displaced a distance x downwards from equilibrium. Take downwards as positive.

a) Write down the tension in the spring when the extension is e.
b) Explain why $mg = \lambda\frac{e}{l}$ when the particle is in equilibrium.
c) When the particle is displaced a further distance x downwards, write down an expression for the extension in the spring.
d) Hence write down an expression for the tension in the spring after this further displacement.
e) When $x > 0$, state the direction of the tension in the spring relative to the equilibrium position. Hence explain why the restoring force is negative.
f) Show that
$$m\frac{d^2x}{dt^2} = mg - \lambda\frac{e+x}{l}.$$
g) Use the equilibrium result to show that
$$m\frac{d^2x}{dt^2} = -\frac{\lambda}{l}x.$$

Question 11. A particle of mass 0.4 kg hangs from a light elastic spring of natural length 0.5 m and modulus of elasticity 8 N. The particle is displaced slightly from equilibrium and released. The spring remains taut. Find the differential equation satisfied by the displacement x from equilibrium at time t seconds.

Question 12. A particle of mass 0.5 kg hangs from a light elastic spring of natural length 0.4 m and modulus of elasticity 20 N. The particle is displaced slightly from its equilibrium position and released from rest. Let x be the downward displacement from equilibrium at time t seconds.

a) Show that the motion satisfies $\dfrac{d^2x}{dt^2} = -100x$.

b) Find x in terms of t, given that the initial displacement is 0.1 m.

Question 13. A particle of mass 0.3 kg hangs from a light elastic spring of natural length 0.5 m and modulus of elasticity 12 N. The particle is displaced 0.08 m downwards from equilibrium and released from rest. Let x be the displacement from equilibrium at time t seconds.

a) Find the differential equation satisfied by x.

b) Find x in terms of t.

Question 14. A particle of mass m kg is attached to a light elastic spring of natural length l and modulus of elasticity λ. The particle performs small vertical oscillations.

a) Show that the motion is simple harmonic and find the angular frequency ω.

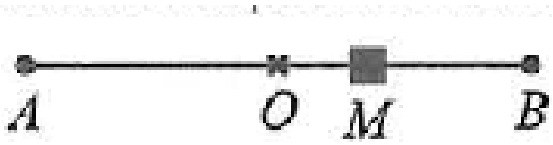
b) Hence find the period of oscillation.

Question 15. A particle of mass m kg moves along a straight line. The force acting on the particle is always directed towards a fixed point O, and has a magnitude proportional to the distance x m at time t s from O. Show that the motion is simple harmonic and hence write down the differential equation of motion.

Question 16. A particle of mass m kg moves along a smooth horizontal surface. At time t_0, where t is the time in seconds, the particle comes into contact with a light spring which is fixed at one end to a rigid wall. At time t, the compression of the spring is x, and the magnitude of the force exerted by the spring on the particle is given by $T = kx$, where k is a positive constant. In a simple model of the motion, the only horizontal force acting on the particle while it is in contact with the spring is this force.

a) Show that, while the particle is in contact with the spring, it performs simple harmonic motion.
b) Find, in terms of k and m, the period of the motion.
c) State the position about which the particle oscillates and explain your answer.

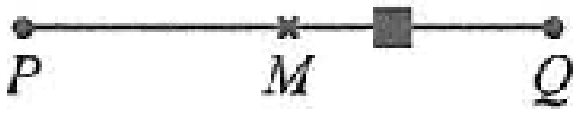
Question 17. A particle of mass M kg is attached between two light elastic springs on a smooth horizontal surface. The left-hand spring is fixed to a point A, and the right-hand spring is fixed to a point B, where $AB = 1.0$ m. Each spring has natural length 0.4 m, and for each spring the magnitude of the tension is given by $T = cx$, where x is the extension of that spring and c is a positive constant. The particle moves along the line AB. The midpoint of AB is O.



At time t seconds, the displacement of the particle from O is x m, where displacement from O towards B is taken to be positive.

a) Show that the resultant force on the particle is $-2cx$ N.
b) Hence show that the particle performs simple harmonic motion.
c) Find the period of the motion.

Question 18. A small trolley of mass 8 kg moves on a smooth horizontal guide. The trolley is attached to two identical light springs. The outer ends of the springs are fixed to points P and Q, which are 1 m apart. Each spring has natural length 0.3 m, and the magnitude of the tension in each spring is given by $T = 160e$, where e metres is the extension in the spring. The trolley moves along the line PQ. The midpoint of PQ is M.

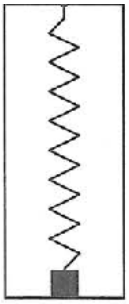


The trolley is released from rest at a point 0.1 m to the right of M. At time t seconds after release, the displacement of the trolley from M is x metres, where displacement to the right is taken as positive.

a) Show that the resultant force on the trolley at time t is $-320x$ N.
b) Hence show that the trolley moves with simple harmonic motion.
c) Find the period of the motion.
d) Find an expression for x in terms of t.
e) Find the speed of the trolley when it is 0.04 m from M.

Q19–Q20 use $g = 9.8$ m s^{-2}.

Question 19. A metal bob of mass 1 kg is attached to a spring of natural length 0.3 m. The magnitude of the tension, T newtons, in the spring is proportional to its extension. When the bob hangs in equilibrium, the length of the spring is 0.5 m. The bob is then placed inside a tall cylinder filled with a viscous liquid, with the spring attached to a fixed point P at the top of the cylinder. The inside depth of the cylinder is 0.8 m.



The bob is released from rest at the base of the cylinder, directly below P. At time t seconds, the displacement of the bob below P is x metres. As the bob moves, it experiences a resistive force of magnitude $14v$ N, where v is the speed of the bob at time t. Assume that the only forces acting on the bob are its weight, the tension in the spring and the resistive force described above.

a) Show that the magnitude of the tension in the spring is given by $T = 49e$, where e is the extension of the spring.

b) Show that the motion of the bob satisfies $\dfrac{d^2x}{dt^2} + 14\dfrac{dx}{dt} + 49x = 24.5$.

c) Find an expression for x in terms of t.

d) Show that 0.8 seconds after being released, the bob has almost reached its equilibrium position.

e) Describe the motion of the bob in the long term. Give a reason for your answer.

Question 20. A light spring is fixed to the underside of a horizontal plate, and a particle of mass m kg is attached to the lower end of the spring. The particle and spring are fully immersed in the viscous fluid.



When the extension of the spring is e metres, the tension in the spring is $16me$ N. The particle is held in rest in a position where the extension of the spring is 0.25 m. The particle is then released from rest. During the subsequent motion, the liquid exerts a resistive force of magnitude $32mv$ N on the particle, where v m s^{-1} is the speed of the particle. At time t seconds after release, the displacement of the particle downwards from its initial position is x metres.

a) Find x in terms of t.

b) State, giving a reason, the type of damping that occurs.

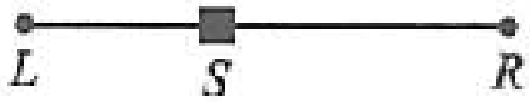
Question 21. A light elastic string has natural length 2 m. The tension in the string is given by $T = 6e$, where e metres is the extension of the string.

a) State the stiffness of the string.

b) Find the tension when the string has length 2.3 m.

c) The string is stretched until the tension is 9 N. Find the length of the string.

Question 22. A small slider S of mass $4d$ kg moves on a smooth horizontal rail. The slider is attached to two light elastic cords. One cord joins S to a fixed point L, and the other joins S to a fixed point R, where L and R are 10 m apart. The points L, S and R always lie on the same horizontal line.



The cord LS has natural length 3 m and stiffness $2d$ N m^{-1}. The cord SR has natural length 2 m and stiffness $6d$ N m^{-1}. The slider is pulled 0.2 m from its equilibrium position towards L and released from rest.

a) Show that the slider moves with simple harmonic motion.

b) Find the angular frequency of the motion.

c) Find the speed of the slider when it is 0.12 m from its equilibrium position.

Prerequisite knowledge for Q23: The Coefficient of Friction

Question 23. A small block Q of mass m kg is attached to two light elastic springs, one joining Q to a fixed point L and the other joining Q to a fixed point R. The points L and R are fixed on a rough horizontal shelf and are 5 m apart. The points L, Q and R always lie on the same straight line.



The string LQ has natural length 1.2 m. When its extension is e m, the tension in the string is $18me$ N. The string QR has natural length 1.8 m. When its extension is e m, the tension in the string is $27me$ N. The coefficient of friction between the block and the shelf is 0.5.

a) Show that when $LQ = 2.4$ m, the tension in string LQ is equal to the tension in string QR.

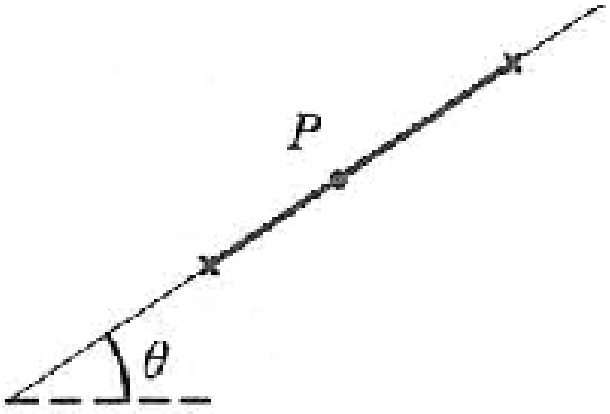
The block is held at a point between L and R where $LQ = 2.8$ m and is then released from rest. At time t seconds after release, the distance LQ is $(2.4 + x)$ m.

b) Show that, while the block is moving towards L, $\dfrac{d^2x}{dt^2} + 45x = 4.9$.

The shelf is now replaced by a smooth horizontal channel containing a viscous liquid. The block is again released from rest from the point where $LQ = 2.8$ m. As the block moves, the liquid exerts a resistive force of magnitude $8mv$ N on the block, where v m s^{-1} is the speed of the block.

c) Find x in terms of t while the block is moving towards L.

Question 24. A particle P with mass m kg moves on a smooth plane inclined at an angle θ to the horizontal. The particle is attached to two light elastic strings which lie along the line of greatest slope of the plane. One string is fixed to a point above P, and the other is fixed to a point below P.



The particle is initially at its equilibrium position. It is then displaced a distance x down the slope. When the particle is at equilibrium, the upper string has extension a m and the lower string has extension b m. The tension in the upper string is k_1e when its extension is e m, and the tension in the lower string is k_2e when its extension is e m. Take down the slope as positive.

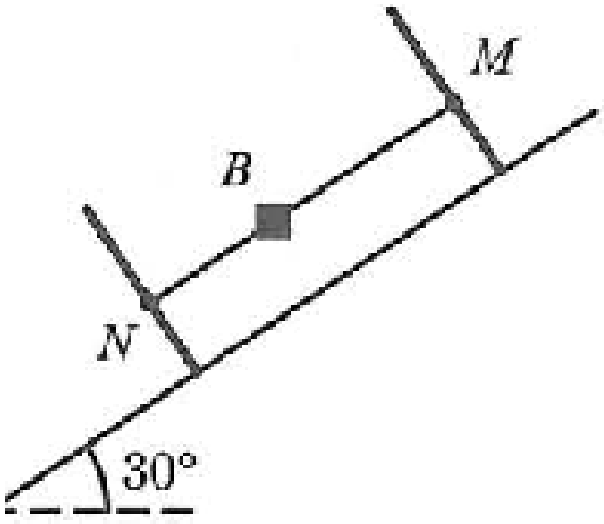
a) Write down the extension of each string after the particle has been displaced a distance x down the slope.

b) State the component of the weight of the particle parallel to the plane.

c) Write down an equation of motion for the particle parallel to the plane.

Q25 use $g = 9.8$ m s^{-2}.

Question 25. A bead, B, of mass 0.5 kg can slide on a smooth straight wire. The wire lies along a line of greatest slope of a fixed plane which is inclined at 30° to the horizontal. Two light elastic strings are attached to the bead. One string joins the bead to a fixed point M higher up the wire, and the other joins the bead to a fixed point N lower down the wire.



The distance MN is 4.2 m. For the string MB, the tension is 8 N when its extension is e m, and its natural length is 1.5 m. For the string BN, the tension is 12 N when its extension is e m, and its natural length is 0.9 m. The bead is pulled 0.15 m down the wire from its equilibrium position and released from rest. At time t seconds after release, the displacement of the bead from equilibrium, measured down the wire, is x m.

a) Show that the motion of the bead is simple harmonic, and that x satisfies $\ddot{x} + 40x = 0$.

A second experiment involves the wire and bead immersed in a viscous fluid. The fluid exerts a resistive force of magnitude cv N on the bead, where v m s^{-1} is the speed of the bead. The motion is critically damped.

b) Show that $c = 2\sqrt{10}$.

c) Find x in terms of t.

Question 26. A particle P of mass 1 kg moves on a smooth plane inclined at 45° to the horizontal. The particle is attached to one end of a light elastic string, whose other end is fixed at a point A higher up the line of greatest slope of the plane. The string has natural length 2 m and modulus of elasticity 20 N. When the particle is moving, a resistive force of magnitude $2v$ N acts on it, where v m s^{-1} is the speed of the particle. The particle is displaced 0.2 m down the slope from its equilibrium position and released from rest. At time t seconds after release, the displacement of the particle from equilibrium, measured down the slope, is x m.

a) Find the extension of the string when the particle is in equilibrium.

b) Show that during the subsequent motion, $\dfrac{d^2x}{dt^2} + 2\dfrac{dx}{dt} + 10x = 0$.

c) Find x in terms of t.

d) Find the displacement of the particle at the first instant after release when it is again momentarily at rest.